

ZHIGALOVO, Russian Federation

WMO: 305210

Lat: 54.800N

Lon: 105.167E

Elev: 419

StdP: 96.39

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.3	(c) -40.0	(d) -44.8	(e) 0.0	(f) -42.1	(g) -42.5	(h) 0.1	(i) -39.8	(j) 7.5	(k) -15.4	(l) 5.6	(m) -16.1	(n) 0.1	(o) 300	(p) 0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	30.3	18.2	28.3	17.4	26.2	16.7	20.0	26.3	19.0	25.0	18.1	23.8	1.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.7	22.0	17.1	12.8	21.3	16.0	11.9	20.2	59.6	26.3	56.1	24.8	53.1	24.0	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.0	(c) 4.7	(e) -43.9	(f) 34.4	(g) 3.1	(h) 1.8	(i) -46.1	(j) 35.6	(k) -47.9	(l) 36.7	(m) -49.6	(n) 37.7	(o) -51.8	(p) 38.9		
(a) 7.4	(b) 6.0	(c) 4.7	(e) -43.5	(f) 21.6	(g) 2.8	(h) 1.3	(i) -45.5	(j) 22.6	(k) -47.1	(l) 23.3	(m) -48.7	(n) 24.1	(o) -50.7	(p) 25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.3	-25.9	-20.9	-9.6	1.3	8.0	15.6	18.7	15.8	7.9	-1.3	-14.3	-24.2	(6)	
(7)		DBStd	16.62	7.67	7.25	6.60	4.63	4.20	3.88	3.01	3.47	4.11	5.28	8.29	8.72	(7)	
(8)		HDD10.0	5174	1113	865	609	263	90	4	0	3	88	350	730	1059	(8)	
(9)		HDD18.3	7610	1372	1098	867	512	320	99	33	92	312	608	980	1317	(9)	
(10)		CDD10.0	680	0	0	0	1	28	172	270	182	27	0	0	0	(10)	
(11)		CDD18.3	73	0	0	0	0	0	16	44	13	0	0	0	0	(11)	
(12)		CDH23.3	1203	0	0	0	0	47	366	549	229	12	0	0	0	(12)	
(13)		CDH26.7	367	0	0	0	0	8	112	184	61	2	0	0	0	(13)	
(14)	Wind	WSAvg	1.4	0.8	1.0	1.5	2.2	2.3	1.7	1.4	1.5	1.5	1.2	0.8	(14)		
(15)	Precipitation	PrecAvg	333	11	7	6	10	21	46	77	73	35	14	16	17	(15)	
(16)		PrecMax	477	30	22	15	28	43	100	193	130	74	39	28	34	(16)	
(17)		PrecMin	242	2	2	0	0	4	7	19	12	7	0	6	6	(17)	
(18)		PrecStd	54	5	4	4	6	9	21	40	29	16	9	6	7	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.1	-0.8	8.9	19.6	28.0	32.8	33.2	31.2	24.8	14.9	3.2	-3.6	(19)	
(20)			MCWB	-5.7	-3.0	2.8	9.5	14.7	17.8	19.4	18.7	15.0	8.2	1.0	-4.5	(20)	
(21)		2%	DB	-8.6	-3.6	5.9	15.7	24.5	29.6	30.9	28.2	20.8	11.0	1.2	-5.7	(21)	
(22)			MCWB	-9.4	-5.1	1.1	7.1	12.8	16.9	19.2	17.8	13.0	6.1	-0.3	-6.5	(22)	
(23)		5%	DB	-11.9	-6.8	3.1	12.5	21.4	27.5	28.7	25.9	17.7	8.1	-0.8	-8.2	(23)	
(24)			MCWB	-12.5	-7.9	-0.6	5.4	11.3	16.3	18.3	16.8	11.6	5.1	-2.0	-8.9	(24)	
(25)		10%	DB	-14.7	-9.7	0.8	9.6	18.0	25.1	26.4	23.5	15.0	5.6	-2.9	-11.4	(25)	
(26)			MCWB	-15.2	-10.6	-2.0	4.0	9.6	15.6	17.8	16.1	10.1	3.1	-3.9	-11.8	(26)	
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.5	-2.6	3.3	9.8	15.3	20.2	22.2	20.4	15.7	9.6	1.8	-4.4	(27)
(28)				MCDWB	-5.2	-1.1	8.3	18.9	25.4	26.4	29.2	28.2	22.2	12.9	2.8	-3.7	(28)
(29)	2%		WB	-9.4	-5.1	1.3	7.7	13.5	18.9	20.7	18.8	13.8	7.0	-0.3	-6.5	(29)	
(30)			MCDWB	-8.6	-3.5	5.4	14.1	22.6	25.6	27.3	25.4	19.3	9.9	0.9	-5.8	(30)	
(31)	5%		WB	-12.4	-8.0	-0.4	6.0	11.7	17.7	19.7	17.8	12.4	5.0	-2.0	-8.8	(31)	
(32)			MCDWB	-11.9	-7.0	2.7	11.8	19.9	24.7	25.4	24.0	16.6	8.0	-0.9	-8.1	(32)	
(33)	10%		WB	-15.2	-10.6	-1.9	4.2	10.2	16.4	18.8	16.7	10.8	3.3	-4.0	-11.9	(33)	
(34)			MCDWB	-14.7	-9.7	0.7	9.1	17.2	23.5	24.2	22.0	14.1	5.4	-3.0	-11.4	(34)	
(35)	Mean Daily Temperature Range		5% DB	MDBR	9.5	14.0	16.4	13.1	14.8	14.6	13.1	12.6	11.8	9.8	9.3	8.8	(35)
(36)				MCDBR	9.7	14.1	17.5	18.5	22.0	19.9	18.7	18.3	17.4	14.2	8.9	8.9	(36)
(37)		MCWBR		9.3	12.4	12.8	10.4	11.0	8.1	7.7	8.1	9.0	9.2	7.6	8.3	(37)	
(38)		5% WB	MCDWB	9.6	13.9	16.3	17.1	19.8	15.8	14.2	15.3	15.6	13.1	8.6	9.1	(38)	
(39)			MCWBR	9.2	12.1	12.0	10.2	10.4	7.1	6.8	7.4	8.4	8.7	7.6	8.5	(39)	
(40)																	(40)
(41)	Clear-Sky Solar Irradiance	taub	0.276	0.305	0.329	0.365	0.406	0.441	0.454	0.394	0.341	0.322	0.294	0.272	(41)		
(42)		taud	2.047	2.034	2.060	2.193	2.205	2.183	2.178	2.337	2.446	2.366	2.143	2.015	(42)		
(43)		Ebn at Noon	628	749	828	854	837	808	788	820	826	746	606	530	(43)		
(44)		Edh at Noon	69	100	124	126	132	137	136	109	85	73	62	57	(44)		
(45)	All-Sky Solar Radiation	RadAvg	0.99	2.01	3.45	4.75	5.30	5.76	5.26	4.49	3.03	1.75	0.94	0.67	(45)		
(46)		RadStd	0.08	0.10	0.17	0.34	0.44	0.39	0.47	0.27	0.28	0.20	0.08	0.07	(46)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	-2.18	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page